

PKCERT AI & Software Development Internship

Task 06 – Exploratory Data Analysis (EDA) Using Jupyter Notebook

Objective:

Develop practical skills in Exploratory Data Analysis (EDA) using Python and Jupyter Notebook by analyzing a real-world dataset, creating visualizations, and presenting written insights.

Total Marks: 100

Part A – Dataset Selection (10 Marks)

1. Select a public dataset from Kaggle, the UCI Machine Learning Repository, or another open data source.
2. Describe the dataset, including its source, purpose, number of records, features, and target variable (if applicable).

Part B – Exploratory Data Analysis (50 Marks)

1. Explore the dataset by examining its structure, data types, and summary statistics.
2. Identify missing values, duplicate records, and inconsistencies.
3. Create at least five meaningful visualizations using Matplotlib and/or Seaborn.
4. Interpret each visualization by explaining the trends, relationships, distributions, or anomalies observed.

Part C – Written Analysis & Insights (25 Marks)

1. Summarize the key findings from your analysis.
2. Identify patterns, correlations, and potential business or research insights.
3. Discuss dataset limitations and suggest possible improvements or future analyses.

Part D – Notebook Organization (15 Marks)

1. Prepare a well-structured Jupyter Notebook.
2. Use markdown cells, headings, and comments to explain each stage of the analysis.
3. Ensure the notebook is clear, organized, and professionally documented.

Evaluation Criteria

Assessment Area

Marks

Dataset Selection	10
Exploratory Data Analysis	50
Written Analysis & Insights	25
Notebook Organization	15